

Tanvir Ahmed Fahim

Cumilla-3501, Bangladesh

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OBJECTIVE

To apply my engineering knowledge and technical skills in a professional organization while contributing effectively and developing my career.

EDUCATION

- **Bangladesh Army International University of Science and Technology** Cumilla, Bangladesh
Bachelor of Science in Electrical and Electronic Engineering (EEE) Oct, 2022 – Oct, 2026
Emphasis: Communication Engineering, RF & Microwave Engineering, Electromagnetics, Electronics
Current CGPA: 3.81/4.00 (Thesis defense pending)
Relevant Coursework: AC & DC Circuit, Energy Conversion, Digital & Analog Electronics, Power System, Communication Theory, Microprocessors and Interfacing, Engineering Electromagnetics, Digital Signal Processing, Solid State Device, Power Electronics, Control Systems, VLSI, Optical Fiber Communication

WORK EXPERIENCE

- **Nagorik TV** Dhaka, Bangladesh
Designation: Industrial Trainee 03/09/2026 – 24/09/2026
 - Gained practical exposure to television broadcasting operations, including studio production, control-room workflows, content management, and transmission systems.
 - Developed an understanding of the complete broadcast workflow across Studio, PCR, MCR, Ingest, Editing, Newsroom, CER, Archive, and RF transmission.
- **Bangladesh Army** Bangladesh
Designation: Soldier Jul 2019 – Apr 2022
 - Served in multiple operational roles, gaining experience in discipline, teamwork, and tactical field operations.
- **Field Intelligence Unit** Cox's Bazar, Bangladesh
Designation: Wireless Operator Oct 2020 – May 2021
 - Served in the Field Intelligence Unit of the Bangladesh Army as a Wireless Operator, operating secure communication systems and supporting intelligence missions.

RESEARCH PROJECTS

- **ESP32-Based Multimodal Health Monitoring System** Jul., 2026
ESP32, Biomedical Instrumentation, IoT, MAX30100, EEG, Blynk
 - Designed and developed an ESP32-based multimodal health monitoring system integrating a MAX30100 pulse-oximeter, three-electrode EEG sensor, and OLED display for real-time physiological-signal acquisition, with Wi-Fi-enabled remote monitoring through the Blynk IoT platform.
- **Advanced Pipe Inspection Robot** Jan, 2025
Control System, Trajectory, Navigation, Comprehensive, Transmission, Adaptive
 - Designed an advanced pipe inspection robot for complex pipeline infrastructures, featuring a wall-press mechanism, a sensor suite, an adaptive control system, and real-time data transmission, offering a robust and scalable solution for modern infrastructure inspection.
- **Smart Car Parking System** Dec, 2024
IoT, Microcontroller, System Design and Setup
 - Developed an IoT-based smart parking system utilizing IR obstacle sensors, servo motors, and LED indicators for automated gate control and real-time occupancy monitoring, reducing manual effort and improving efficiency.

- **IoT-Based Transmission Line Fault Detection System for Bangladesh's Power Grid** Oct, 2024
GPS, GSM, Fault Detection, Transmission Line
 - Designed an IoT-enabled system for detecting and locating transmission line faults using real-time sensor data, GPS for fault localization, GSM for instant alerts, and Wi-Fi for remote monitoring.
- **Design and Implementation of FIR Filters for Image Processing Applications** Aug, 2024
Digital Signal Processing, MATLAB, FIR Filter Design
 - Implemented FIR filter design techniques to enhance edge detection and reduce noise in image processing applications, improving accuracy in medical imaging diagnostics and security surveillance systems.

PUBLICATIONS

1. Md. N. Alam, Md. Z. Hasan Buiyan, Md. A. Al Hasan Anik, F. Tanvir, and **T. A. Fahim**, "Enhancing Power Grid Reliability in Bangladesh: An IoT-Based Approach to Transmission Line Fault Detection," in *2024 IEEE 3rd International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh: IEEE, Nov. 2024, pp. 224–229. DOI:10.1109/RAAICON64172.2024.10928383.
2. Md. A.-A.-Mamun, **T. A. Fahim**, J. K. Dea, Z. H. Buiyan, A. Rahman, N. M. M. Mujahid, and S. A. Bipul, "Grid-Slot Hybrid Structure Wideband Aperture Coupled Microstrip Antenna Design for C-Band Applications," *Journal of Electrical Systems*, Dec. 18, 2024. Available: <https://journal.esrgroups.org/jes/article/view/7502>.
3. **T. A. Fahim**, Md. S. Rana, N. M. M. Mujahid, and T. Hasan, "28 GHz Patch Antennas for 5G Wireless Applications: A Comprehensive Review," in *Information Systems Engineering and Management*, 2026, pp. 205–218. DOI: 10.1007/978-3-032-19196-0₁₆.
4. **T. A. Fahim**, Md. A.-Mamun, S. A. Bipul, T. Hasan, and H. Datta, "Design of a High Performance Nano Band Microstrip Antenna with Three Rectangular Side Slots for THz Applications," *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026)*. DOI: 10.1109/QPAIN69676.2026.11545910.
5. **T. A. Fahim**, Md. A.-Al-Mamun, J. K. Dea and Md. G. Mustafa, "Machine Learning-Driven Gain Prediction of a High-Performance Dual-Wideband Decoupling-Enhanced Graphene MIMO Antenna for THz Applications." *Diamond & Related Materials*. DOI: 10.1016/j.diamond.2026.114121.
6. **T. A. Fahim**, Md. S. Rana, S. A. Bipul, T. Hasan, N. M. M. Mujahid, and H. Datta, "Design of a Compact Modified Inset-Fed Circular and Inverted U-Shaped Patch Antenna for Terahertz Applications," *11th International Conference on Information and Communication Technology for Intelligent Systems (ICTIS 2026)*. (Accepted)
7. Md. A. Haque, **T. A. Fahim**, M. H. Riad, J.-J. Tiang, and N. S. S. Singh, "Compact THz MIMO Antenna with Multi-Resonance and Low Mutual Coupling for High-Speed Communication," *Engineering, Technology & Applied Science Research*. (Accepted)
8. Md. A. Haque, **T. A. Fahim**, M. H. Riad, Md. S. Mahmud, J.-J. Tiang, and N. S. S. Singh, "High-Performance Dual-Band Graphene-Based THz MIMO Antenna with Excellent Diversity Characteristics for 6G Wireless Systems," *Engineering, Technology & Applied Science Research*. (Accepted)
9. **T. A. Fahim**, Md. A.-Al-Mamun, J. K. Dea, and N. J. Sumaiya, "A Dual-Band High-Gain Multiresonance Graphene Based THz Antenna for Future 6G Applications," *2026 IEEE International Conference on Microwave, Antennas, RF and Sensors*. (Accepted)
10. **T. A. Fahim**, Md. A.-Al-Mamun, Md. G. Mustafa, and Md. A. Haque, "Isolation-Wall Integrated Graphene THz MIMO Antenna with Wideband and High-Diversity Performance for 6G Applications," *2nd International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS 2026)*. (Accepted)
11. **T. A. Fahim**, M. H. Riad, J.-J. Tiang, N. S. S. Singh, and Md. A. Haque, "Design and Performance Analysis of a PBG Substrate MIMO Antenna with RLC Validation for 6G and IoT Applications," *Discover Internet of Things*. (Under Review)

12. **T. A. Fahim**, M. H. Riad, J.–J. Tiang, and N. S. S. Singh, Md. A. Haque “A Clustered Circular-Slot Graphene THz MIMO Antenna with RLC-Validated Hybrid Learning for UAV-Enabled 6G Smart City,” *Scientific Reports*. (Under Review)
13. Md. A. Haque, **T. A. Fahim**, M. H. Riad, J.–J. Tiang, N. S. S. Singh, Md. M. Islam, M. R. Rashel, and M. Tlemcani, “ML/ DL Assisted Design of a Flower-Shaped Two-Port Antenna for 6G IoT Applications with ECM Validation,” *Engineering Reports*. (Under Review)
14. **T. A. Fahim**, Md. A.–Al-Mamun, J. K. Dea and Md. G. Mustafa, “Compact Multiresonant Graphene-Based THz Antenna with RLC Circuit Validation and ML-Assisted Efficiency Prediction for Advanced Wireless Applications,” *Scientific Reports*. (Under Review)
15. Md. A. Haque, **T. A. Fahim**, M. H. Riad, J.–J. Tiang, and N. S. S. Singh, “Design and Multi-Level Validation of a Penta-Slot Graphene THz MIMO Antenna Using RLC Circuit Modeling and Hybrid Machine Learning,” *SN Computer Science*. (Under Review)

ACHIEVEMENTS

• Best Poster Award, BAIUST Inter Department Poster Presentation Competition	<i>Dec, 2025</i>
• Best Soldier Award–2021 (9 Signals Battalion)	<i>Dec, 2021</i>
• 10%–40% Waiver for Achieving VC List Certificates (1st–4th,6th–7th semester)	<i>2022–2026</i>
• Scholarship on Board of Secondary School Certificate–2016	<i>Feb, 2016</i>

LANGUAGES AND TECHNICAL SKILLS

Human Languages: Bengali (Native), English (Proficient in all aspects)
Simulation & EM Tools: HFSS, CST, COMSOL, ADS
Programming: Python, C, Arduino, MATLAB
Circuit & VLSI Design: LTSpice, PSpice, Proteus, PSCAD, Microwind
3D Design: SolidWorks, AutoCAD
Others: Simulink, Visio, LaTeX, Origin, SigmaPlot

COMMUNITY SERVICE & VOLUNTEERING ACTIVITIES

• IEEE BAIUST Student Branch — Senior Executive Member	<i>Nov, 2024 – Nov, 2025</i>
• BAIUST Robotics Society — General Secretary	<i>Sep, 2024 – Sep, 2025</i>
• Shaheen Academy Schoool And College Scouts — Member	<i>Jan, 2011 – Jan, 2016</i>
• Bangladesh National Cadet Corps — Cadet	<i>Jun, 2016 – Dec, 2017</i>

REFEREES

Dr. Mohammad Shafiul Alam Associate Professor & Head, Department of EEE Bangladesh Army International University of Science and Technology, Cumilla Email: shafiul.eee@baiust.ac.bd Mobile: +880 1977-181410	Md. Abdullah-Al-Mamun Assistant Professor, Department of EEE Bangladesh Army International University of Science and Technology, Cumilla Email: mamun.eee@baiust.ac.bd Phone: +880 1701-012428
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